



**BACHELOR OF INFORMATION TECHNOLOGY (HONS)
BACHELOR OF COMPUTER SCIENCE (HONS)**

MAY 2026 FINAL ONLINE EXAMINATION

COURSE: DATABASE FUNDAMENTALS

COURSE CODE: BTL2233/ BCL2233/ BIT1223

TIME ALLOWED: 2 HOURS

DATE: __AUGUST 2026

This examination question paper consists of FIVE (5) pages including the cover page.

INSTRUCTIONS TO CANDIDATE:

1. Answer all **FOUR (4)** questions.
2. All questions carry equal marks.
3. This is a **closed-book** examination. No books, notes, cell phones, communication devices of any sort, or other aids are permitted except as stated otherwise.
4. All answers must be written in the answer booklet provided.

ADDITIONAL INFORMATION:

1. You are allowed to use a non-programmable calculator.

Answer all FOUR (4) questions.

QUESTION 1

Given a table schema as shown in Table_Q, answer all questions.

Table	Attribute	Data Type	Constraints
SALESPERSON	SalespersonNo SalespersonName Commission hireDate	char (4) Varchar (15) Number date	Primary Key
PRODUCT	productNo productName unitPrice	char (6) Varchar (20) Number (8,2)	Primary Key
SALES	salespPersonNo productNo quantity	char (4) char (6) number	Primary Key Primary Key

Table Q1

a) Write a creation script for all **THREE (3)** tables.

(15 marks)

b) Write insert script for 2 records for ANY **ONE (1)** table.

(10 marks)

(TOTAL 25 Marks)

QUESTION 2

Given a table schema as shown in Table_Q2, answer all questions related to it.

Table	Attribute	Data Type	Constraints
SALESPERSON	SalespersonNo SalespersonName Commission hireDate	char (4) Varchar (15) Number date	Primary Key
PRODUCT	productNo productName unitPrice	char (6) Varchar (20) Number (8,2)	Primary Key
SALES	salespPersonNo productNo quantity	char (4) char (6) number	Primary Key Primary Key

Table Q2

- a) Write a SQL statement to display all salesperson sort by hire date.
(3 marks)
- b) Write a SQL statement to display all salespersons with commission more than 10%.
(4 marks)
- c) Write a SQL statement to display all products' unit prices after 10% increase.
(4 marks)
- d) Write a SQL command to display all salespersons, product number and quantity for products wrench or hammer, sort the result by salesperson.
(6 marks)
- e) Write a SQL command to display salesperson number, salesperson name, product name and total for each salesperson sold for each product.
(8 marks)

(Total: 25 marks)

QUESTION 3

- a) Given an Order Form in Figure 1 below, draw a complete ERD(Entity Relationship Diagram) to store the data.

Note: Must consist of entity, attributes, primary key, cardinality, junction table.

Order Form			
Order number: 1234 Date: 11/04/98			
Customer number: 9876			
Customer name: Billy			
Customer address: 456 HighTower Street			
City-Country: Hong Kong, China			
ProductNo	Desscription	Quantity	Unit Price
A123	Pencil	100	\$3.00
B234	Eraser	200	\$1.50
C345	Sharpener	5	\$8.00

Figure 1: Order Form

(15 marks)

- b) Based on the ERD designed in Question 2 (a), write a complete SQL creation script using Oracle syntax to create **ONE (1)** table (any table). This should include table name, data type, column constraint and appropriate symbols.

(10 marks)

(Total: 25 marks)

QUESTION 4

Shown in Figure Q3 below is a Production Request Form. Answer the questions below.

Form 2: Production Request Form

Production Request PR No. 					
Supplier No Supplier Name Date of order Signature Expected Delivery Date			Supplier Contact Info:		
Supplier Item #	Item Description	Unit Size	Unit price	Quantity	Sub Total

Figure Q3: Production Request Form

- a) Normalize the Production Request Form from Unnormalized Form (UNF) to Third Normal Form (3NF). Clearly show each stage of the normalization process.
(10 marks)
- b) Describe **THREE (3)** database anomalies and provide examples for each.
(15 marks)
(Total: 25 marks)

- END OF QUESTION PAPER -